

Endogenous Merger Decisions Among Competitors: Impact of Limited Capacity and Loyal Segments

Derui Wang,^a Xiaole Wu,^{b,*} Christopher S. Tang,^c Yue Dai^b

^aInternational Institute of Finance, School of Management, University of Science and Technology of China, Hefei, Anhui 230026, China;

^bSchool of Management, Fudan University, Shanghai 200437, China; ^cAnderson School of Management, University of California, Los Angeles, Los Angeles, California 90095

*Corresponding author

Contact: wangderui@ustc.edu.cn,  <https://orcid.org/0000-0001-9003-7525> (DW); wuxiaole@fudan.edu.cn,

 <https://orcid.org/0000-0003-0809-1843> (XW); chris.tang@anderson.ucla.edu,  <https://orcid.org/0000-0001-9597-7620> (CST); yuedai@fudan.edu.cn,  <https://orcid.org/0000-0001-5964-6509> (YD)

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Abstract. We study endogenous merger decisions among three competing firms with asymmetric capacity: one large firm has ample capacity, and two small firms have limited capacity. Each firm can either operate independently or merge with another firm. Because of antitrust laws, all three firms merging is not permitted. When two firms merge, the merged entity and the remaining independent firm use the resulting capacity to capture their own “loyal buyers” and compete in price for price-sensitive “disloyal switchers.” We first consider a base model in which the size of loyal buyers for different firms is symmetric. We find that, in equilibrium, either there is no merger or the large firm will merge with a small firm as a “mixed merger.” We explain how this equilibrium structure is driven by the mergers’ impacts on market competition and firms’ pricing aggressiveness. Specifically, the mixed merger that can most effectively reduce price competition is the equilibrium under relatively competitive market conditions (i.e., the size of loyal buyers is small and capacity is large but limited). We also extend to the case with asymmetric size of loyal segments and show that two small firms merging (“small merger” structure) can be the equilibrium structure when the large firm has a much larger loyal segment than the small firms and the small firms’ capacity is sufficiently large. The impacts of mergers on social welfare and consumer surplus provide practical implications from the antitrust perspective. The antitrust agencies can approve a mixed merger to improve social welfare under certain conditions, but if the agencies pay more attention to consumer surplus, then mixed mergers should not be permitted. Small mergers, though sometimes not the equilibrium structures, can improve social welfare and consumer surplus. In such cases, the government can appropriately promote mergers that are beneficial for society and/or consumers by adopting some restrictive measures or offering incentives.

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1. Introduction

To increase market share and reduce market competition, companies (within the same industry) often propose mergers that involve large and small firms. For instance, in 2011, the leader in analog semiconductors, Texas Instruments (TI), acquired a small rival, National Semiconductor. In the same vein, the second largest firm, Analog Devices with a 9% market share, completed the acquisition of Maxim with a 4% market share in 2021. In 2022, the largest integrated device manufacturer (IDM), Intel, acquired Tower Semiconductor, which is a small foundry.¹

The above merger examples share three common salient features: (i) competition—the merging firms were direct competitors before the merger, (ii) capacity—the merging firms have substitutable capacity (e.g., chip fabs for semiconductor manufacturers are substitutable when the fabs are used to produce chips with the same node size), and (iii) loyal customers—each firm has its own loyal customers. However, we notice that the features of capacity and loyal buyers have not been considered much in the merger literature so that their impacts on mergers remain unclear. Specifically, researchers primarily focus on evaluating factors

(e.g., complementarity, risk pooling, bargaining power) that drive noncompeting firms with complementary capacity to form coalitions (Nagarajan and Sošić 2009). In business practice, sizes of loyal customers and capacity are two significant factors in merger decisions among competitors.

One of key features in the semiconductor manufacturing industry is that it usually takes years to increase capacity by building new fabs (Blum 2021). Therefore, to expand production capacity quickly, some manufacturers choose to acquire competitors. In the announcement of the merger between TI and National Semiconductor, TI explicitly states that it could benefit from National's manufacturing capacity (Texas Instruments 2011a, b). Intel also stated that the acquisition of Tower Semiconductor could further expand its manufacturing capacity to advance its goal of becoming a major provider of foundry capacity globally (Intel 2022). The capacity constraint has been regarded as an essential factor in mergers but has not been well studied (Cho 2014, Cabolis et al. 2021).

Another key feature is that, when a buyer changes its semiconductor manufacturer, a new product development process is needed, and this takes at least 1.5 years (Burkacky et al. 2021). The reason is that the manufacturing process and manufacturing intellectual property are manufacturer-specific, so alterations in the chip design are needed to meet these requirements. Therefore, it is not easy for some semiconductor buyers to switch suppliers, and these buyers can be interpreted approximately as loyal buyers. As suggested by a report of the World Economic Forum, "Semiconductor companies could undertake more M&A [mergers and acquisitions] deals to expand their customer base" (Aboagye et al. 2022, p. 8). The realistic need and the research gap motivate us to examine the impacts of substitutable capacity and loyal segments on merger decisions among competing firms.

As an initial attempt, we consider potential mergers of three competing firms (one large and two small firms) with asymmetric capacity and either symmetric or asymmetric size of loyal buyers. Note that only a merger between two firms is permissible because a monopolistic merger of three firms is not allowed because of antitrust laws. We also examine the impacts of mergers on social welfare and consumer surplus to provide antitrust agencies with implications on whether to approve or reject a merger. We aim to answer four research questions: (i) When will the large firm merge with a small firm (i.e., a mixed merger) to compete with the remaining small firm? (ii) Under what conditions will the two small firms merge (i.e., a small merger) to compete with the large firm? (iii) How will each firm's capacity and loyal buyers affect the equilibrium merger structure? (iv) What are the practical implications from the antitrust perspective?

To address the above research questions, we examine potential mergers among three price-setting firms. We use the strong Nash equilibrium concept (Aumann

1959) to identify stable merger structures. Specifically, in an equilibrium merger structure, neither individual firms nor groups of firms can be strictly better off by deviating. We first examine the base model in which all three firms have an identical size of loyal buyers (even though the large firm has ample capacity and small firms' capacity is limited). We extend our analysis to the case when the loyal segment size is firm-specific.

Our analysis identifies two driving forces that influence the firms' equilibrium merger decisions. First, from the perspective of market competition, similar to the result of Deneckere and Davidson (1985) without the consideration of firms' capacity, we find that mergers can decrease the number of competitors and thereby reduce the intensity of competition. Considering that merging firms pool together their capacities, which sometimes enhances competitors' asymmetry in capacity, we further investigate how mergers influence market competition by altering firms' capacity situation. Price competition among firms primarily targets disloyal switchers, and an increase in capacity asymmetry facilitates differentiation in firms' ability to serve switchers, thus dampening market competition. This is a positive driver for firms to merge and is referred to as the asymmetric capacity effect of a merger. The market after a mixed merger between the large firm and a small firm is the most asymmetric among all possible market structures because the size difference between the merged entity and the remaining small firm is the greatest. Hence, the mixed merger dampens the market competition more significantly by generating the most significant asymmetric capacity effect.

Second, from the perspective of firms' pricing aggressiveness, the concentration of loyal buyers makes the merging firms more concerned about the potential profit loss because of price reduction, leading them to become more conservative in cutting prices. This reduction in merging firms' pricing aggressiveness increases the likelihood that merging firms charge higher prices than the stand-alone firm, in which case the merging firms only capture those switchers not able to purchase from the stand-alone firm, which, hence, leads to a decrease in their expected demand and the benefit they might gain from the reduced competition. This is a negative driver for merging.

Our analysis shows that, when all three firms have the same size of loyal buyers, either no merger emerges or the large firm merges with a small firm under conditions pertaining to the size of firms' capacity and loyal segment. Specifically, the mixed merger structure dominates when the initial market competition is intense (i.e., the size of loyal buyers is small and capacity is large but limited) because the mixed merger associates with strong asymmetric capacity effect and can dampen price competition more effectively compared with the small merger. When the initial market competition is not

intense, there is no merger because the positive asymmetric capacity effect is not significant and cannot offset the negative effect of mergers. The small merger structure does not emerge in equilibrium because it does not generate the positive asymmetric capacity effect.

We further extend our base model to the case when the large firm has a larger size of loyal buyers than small firms. In this case, there are two possible price competition patterns in the independent structure: either all three firms compete with each other or only two small firms compete and the large firm chooses not to compete with them by setting the highest possible price to meet only the demand of its loyal buyers. For the latter pattern, capacities of two competing small firms are symmetric, but after the small merger, the two competing parties, the large firm and the merged entity by small firms, have asymmetric capacities. Hence, the small merger can also generate the asymmetric capacity effect to significantly reduce market competition. As a result, the small merger can emerge as the equilibrium on some occasions.

Mergers among competitors are standard cases for anti-trust investigation. In most countries, the objective of anti-trust policies is to protect consumers, and mergers that can enhance market power and substantially lessen market competition should not be permitted (see the merger guidelines: U.S. Department of Justice and Federal Trade Commission 2010, UK Competition and Markets Authority 2021). However, in a conventional economic view, the proper goal of antitrust policy is to maximize overall market efficiency or welfare (Farrell and Shapiro 1990). To provide the antitrust government agency with policy suggestions on how to evaluate the proposed mergers, we investigate the impacts of mergers on both consumer surplus and social welfare. We find that, if small firms have relatively low capacity and few loyal buyers or small firms' capacity is sufficiently high but limited, the merger between a large firm and a small firm can increase social welfare, and meanwhile, the corresponding structure can emerge as the equilibrium. In this case, the antitrust agencies can approve this merger to improve social welfare. However, the agencies should not permit this merger when paying attention to consumer surplus because such mixed mergers typically decrease consumer surplus. Furthermore, the government can explicitly prohibit mergers with a large firm to promote the merger between small firms when the small merger is beneficial for society and/or consumers.

This paper is organized as follows. Section 2 reviews the related literature, and Section 3 presents our model along with the conditions for each equilibrium merger structure. To explicate our analysis, Section 4 considers the benchmark case when firms are not capacity constrained. For the capacity-constrained case, the firms' equilibrium profits are derived in Section 5, and the equilibrium merger structures are examined in Section 6. Section 7 examines the impacts of mergers on consumer

surplus and social welfare. An extension model with asymmetric size of loyal buyers is presented in Section 8. We conclude in Section 9. All tables and some figures are relegated to Online Appendix A. The derivation of equilibrium conditions is provided in Online Appendix B. Some extensions of our paper are analyzed in Online Appendices C–G. All proofs are given in Online Appendix H.

2. Literature Review

Collaboration of noncompeting firms with complementary resources has been widely studied. A stream of literature examines coalition formation in a variety of supply chain structures. For instance, Granot and Yin (2008) investigate push-and-pull contracts between an assembler and its suppliers. The suppliers are complementary to each other and may form coalitions among themselves. They show that suppliers in a pull system can more easily coordinate their production quantities than in a push system. Unlike the model of Granot and Yin (2008) that assumes a fixed retail price and stochastic demand, Nagarajan and Sošić (2009) and Yin (2010) both investigate alliance formation in a decentralized assembly system with price-sensitive demand. In order to capture the effect of channel power on coalition stability, Nagarajan and Sošić (2009) consider multiple farsighted suppliers selling complementary components to a downstream assembler and predict the structure of possible supplier coalitions under different market conditions. Yin (2010) examines how demand characteristics and sequential decision making affect alliance formation. Contracting issues of horizontal alliances formed by noncompeting firms have also received some attention recently. Fang and Cho (2014) study the stability and formation of inventory-transshipment coalitions among noncompeting firms that face demand uncertainty. Roels and Tang (2017) examine capacity allocation contracts between two firms sharing complementary resources. Levi et al. (2020) study strategic capacity investment among multiple firms forming a joint venture and sharing revenue via fixed-rate contracts. Our model differs from these studies in that we consider mergers among three competing firms with asymmetric capacity and focus on analyzing the impacts of capacity and loyal buyers on merger partner selection.

With given collaboration partners, one stream of literature studies the impact of competing firms' collaboration decisions and contract formats on their profits and/or social welfare. Hu et al. (2013) analyze the revenue-sharing scheme for airline alliances in which carriers negotiate static proration rates to split revenues generated from code-sharing and interline itineraries. Ahn et al. (2024) analyze two competing firms' collaboration by either building capacity together or sharing the existing capacity for production. They find that firms

competing in the production stage build less overall capacity than if they coordinate their capacity investment and production decisions.

Our paper is closely related to the literature on horizontal mergers among competitors. The seminal work of Salant et al. (1983) considers a Cournot model in which symmetric firms with constant marginal production costs sell homogenous products. They assume that a merged entity behaves as a multiplant Cournot player and find that exogenous mergers may reduce the endogenous joint profits of the firms that are assumed to collude. Szidarovszky and Yakowitz (1982) and Davidson and Deneckere (1984) also observe this result. By considering a similar quantity-setting game, Perry and Porter (1985) find that firms have incentives to merge when the production cost is quadratic. Farrell and Shapiro (1990) also use a Cournot model but with general cost and demand functions. They conclude that mergers typically raise prices unless they generate considerable synergies or scale economies. There are some other papers considering price-setting games like our paper. Deneckere and Davidson (1985) consider a Bertrand model in which firms produce differentiated products. Different from the result in Salant et al. (1983), Deneckere and Davidson (1985) find that horizontal mergers of any size can increase the joint profits of the participating firms and large mergers yield higher profits than smaller ones. More recently, Moraga-González and Petrikaitė (2013) consider a Bertrand competition model in which firms sell horizontally differentiated products and consumers sequentially search for products. Because the merged entity effectively lowers the consumers' search costs for satisfactory goods at the merging firms, they find that firms can benefit from merging, and both consumer surplus and social welfare may increase after a merger. Federgruen and Hu (2016) study the impact of changes in the network structure and conclude that horizontal mergers may decrease the total industry profits. Cho and Wang (2017) introduce demand uncertainty to this field. They consider three effects caused by a merger between price-setting firms: the collusion effect, the pooling effect, and the synergy effect. Similar to our paper, Sinityn (2020) uses a model with two consumer segments, loyal consumers and switchers, to study the effects of a horizontal merger on consumer surplus and firms' price promotions and profits. These papers all focus on the impact of exogenous mergers. In contrast, we focus on firms' partner selection decisions and the stability of endogenous mergers. In addition, we examine a new supply-side factor: capacity. As emphasized in Perry and Porter (1985), to conceptually model mergers, it is important to show that the merged entity should have access to the combined productive capacity of both merger partners. So far as we know, our paper is the first work studying the impact of firms' capacity on endogenous mergers.

The endogenous alliance formation among competing firms has received much attention in the operations literature. For example, Jin and Wu (2006) investigate auction models in which coalitions could be formed among suppliers to incentivize them to participate in the auctions and enhance their profitability. Granot and Sošić (2005), Sošić (2006), and Nagarajan and Sošić (2007) all focus on alliance formation among farsighted retailers by defining a stable coalition structure according to the largest consistent set (Chwe 1994). Granot and Sošić (2005) analyze the impact of product substitutability and compatibility of retailers on alliance formation, in which the compatibility of retailers is reflected through the relative decreases in wholesale prices stemming from alliance memberships. Similarly, assuming linear demand functions, Nagarajan and Sošić (2007) examine coalition formation of multiple farsighted retailers selling substitutable products. Sošić (2006) analyzes a transshipment setting in which farsighted retailers distribute the profit from inventory sharing according to an allocation rule and shows that the grand coalition could be stable. The growing literature on endogenous mergers is still small. Horn and Persson (2001) use cooperative games to analyze endogenous mergers by defining the dominance relation between market structures via a decisive group of owners. Qiu and Zhou (2007) study the dynamic process of endogenous merger formation. By assuming that firms have asymmetric marginal costs and the merged entity of two firms will shut down the high-cost unit and use the low-cost unit for production, they identify firm heterogeneity and negative demand shocks as two necessary conditions for mergers to occur.

Our work differs from the above studies in several dimensions. First, the purpose and model setting of our research depart substantially from the aforementioned studies. Specifically, we study three competing firms with asymmetric capacity and loyal buyers with a focus on whether the firms will merge in equilibrium, whether the small firms merge or the big firm merges with a small firm, and how capacity and loyal buyer size affect endogenous mergers. Second, our paper differs from the above studies in terms of the definition of the equilibrium merger structure. Following the strong Nash equilibrium concept,² we define a merger structure as an equilibrium when neither individual firms nor groups of firms can be better off by deviating from the market structure. Third, existing studies typically define specific revenue allocation rules within the merged entity, whereas we allow general revenue allocation among merging firms.

3. Model Preliminaries

Consider three competing firms (A, B, and C) that sell perfectly substitutable products (or services) at price p_i , $i = A, B, C$, to potential buyers with identical reservation

value r . For ease of exposition, we scale the size of the entire buyer market to one, normalize each buyer's demand to one unit, and scale each firm's marginal cost to zero. We consider a full-information setting in which each firm's capacity and size of loyal buyers are common knowledge to all firms.

Loyal Buyers and Disloyal Switchers: Before mergers, each of the three firms has the same size of α loyal buyers, $0 \leq \alpha < 1/3$. (We consider the case when the size of loyal buyers varies across firms in Section 8.) These firm-specific loyal buyers buy from firm i as long as $p_i \leq r$, but they never purchase from the other two firms. Besides those 3α firm-specific loyal buyers, the remaining $(1 - 3\alpha)$ buyers are disloyal switchers who buy from the firm that offers the lowest price.³ The loyal buyers are similar to the uninformed ones in Varian (1980) and the loyal ones in Narasimhan (1988). Also, the disloyal switchers are akin to the informed buyers in Varian (1980), switchers in Narasimhan (1988), and shoppers in Kuksov and Zia (2021).

Firm Capacity: To capture capacity asymmetry, we assume firm A is a large firm with an endowed capacity k_A and firms B and C are two small firms with identical capacity so that $k_B = k_C \equiv k < k_A$. Before mergers, all three firms compete for buyers as follows. First, if firm A's price p_A is the lowest and $p_A \leq r$, then firm A's α loyal buyers and all $(1 - 3\alpha)$ disloyal switchers prefer to buy from firm A so that firm A's total demand is $(1 - 2\alpha)$. To ensure that the large firm A is not capacity constrained, assume $k_A \geq (1 - 2\alpha)$. Furthermore, we assume $k > \alpha$ so that each small firm's endowed capacity exceeds the demand of its loyal buyers.⁴ In the event in which a small firm's capacity cannot meet the potential demand, we specify how the capacity is allocated among the buyers later.

Merger Structure: Because of antitrust laws, a monopolistic three-firm merger is not permissible. We consider three potential structures to be determined by three firms endogenously as follows. First, if there is no merger, firms can operate as three independent firms under the independent structure $\{A, B, C\}$. Second, if the large firm A merges with a small firm, say firm B, the structure is denoted as the mixed merger structure, $\{AB, C\}$. Because firms B and C are symmetric, structures $\{AB, C\}$ and $\{AC, B\}$ are essentially the same, and it suffices to restrict our attention to $\{AB, C\}$. Third, if the two small firms merge, we denote the corresponding structure as the small merger structure, $\{A, BC\}$. Once two firms merge, the two merging firms act as a single entity by pooling together their capacities and loyal buyers.

Equilibrium Profits: For the independent structure $\{A, B, C\}$, let $\Pi_i^{(0)}$ denote the equilibrium profit of firm i , $i = A, B, C$, where the superscript (0) represents the independent structure. For the mixed merger structure $\{AB, C\}$, let $\Pi^{(AB)}$ denote the equilibrium profit of the

merged entity formed by firms A and B⁵ and $\Pi_C^{(AB)}$ be firm C's equilibrium profit in this structure. Finally, for the small merger structure $\{A, BC\}$, let $\Pi^{(BC)}$ denote the equilibrium profit of the merged entity formed by firms B and C and $\Pi_A^{(BC)}$ be firm A's equilibrium profit in this structure; because firms B and C are symmetric in capacity, we assume each firm obtains half of the merged entity's profit, that is, $\Pi_B^{(BC)} = \Pi_C^{(BC)} = \Pi^{(BC)}/2$.

3.1. Conditions for an Equilibrium Merger Structure

Because of antitrust laws and the fact that firms B and C are symmetric, there are only three potential structures to consider: independent $\{A, B, C\}$, mixed merger $\{AB, C\}$, and small merger $\{A, BC\}$. We use the solution concept of strong Nash equilibrium. Under the Nash equilibrium, only unilateral deviations by individual players are prevented. When there are more than two players and players can freely discuss their strategies, it is natural to also consider multilateral deviations by a group of players. For this reason, the strong Nash equilibrium refines the concept of Nash equilibrium by also considering multilateral deviations. We now discuss the conditions under which a given structure is an equilibrium structure so that neither individuals nor groups of firms can be strictly better off by deviating from the structure. Three firms are independent initially, as in structure $\{A, B, C\}$, and there is an arbitrarily small fixed transition fee to merge. We now establish these conditions.

3.1.1. Conditions for $\{AB, C\}$ as an Equilibrium Merger Structure. For the mixed merger $\{AB, C\}$ to be an equilibrium structure, we need to establish conditions to prevent two types of deviations associated with the structure $\{AB, C\}$: (i) individual or groups of firms deviate to be independent and (ii) a deviation by firms B and C to merge into one entity. Conditions (1a) and (1b) rule out these two possible deviations so that the mixed merger $\{AB, C\}$ is sustained.⁶

$$\Pi^{(AB)} > \Pi_A^{(0)} + \Pi_B^{(0)}, \quad (1a)$$

$$\Pi^{(AB)} > \Pi_A^{(BC)} + \Pi_B^{(BC)} \text{ or } \Pi_C^{(AB)} \geq \Pi_C^{(BC)}. \quad (1b)$$

Specifically, to prevent deviation (i), we need to establish the condition that ensures neither firm A, firm B, nor they together have an incentive to be independent. When Condition (1a) holds, merged entity AB's profit is greater than the sum of firm A's and firm B's when they are independent. Hence, there is no economic incentive for AB to deviate to be independent.⁷ Furthermore, because Condition (1a) requires $\Pi^{(AB)}$ to be strictly higher than $\Pi_A^{(0)} + \Pi_B^{(0)}$, Condition (1a) not only prevents deviation (i), but also ensures that firms A and B have incentives to merge by deviating from $\{A, B, C\}$ though there is an arbitrarily small fixed transition fee to merge. Next, we consider deviation (ii). Condition (1b) ensures

that firm B has no incentive to deviate and merge with firm C.⁸ To elaborate, if firm B merges with firm C, it earns $\Pi_B^{(BC)}$ and firm A earns $\Pi_A^{(BC)}$. However, to sustain merger AB, firm A can offer firm B a profit of $\Pi_B^{(BC)}$ and keep for itself a profit of $(\Pi^{(AB)} - \Pi_B^{(BC)})$, which is higher than $\Pi_A^{(BC)}$ according to the first condition in (1b). Moreover, firm B's deviation can also be prevented if firm C has no incentive to merge with firm B. The second condition in (1b) guarantees this.

3.1.2. Conditions for $\{A, BC\}$ as an Equilibrium Merger Structure. Conditions (2a) and (2b) ensure that the small merger structure $\{A, BC\}$ is sustained in equilibrium, where

$$\Pi_C^{(BC)} > \Pi_C^{(0)}, \quad (2a)$$

$$\Pi_A^{(BC)} + \Pi_B^{(BC)} \geq \Pi^{(AB)}. \quad (2b)$$

We consider two types of deviations associated with structure $\{A, BC\}$: (iii) individual or groups of firms deviate to be independent and (iv) a deviation by firms A and B to merge into one entity. Condition (2a) (which also implies $\Pi_B^{(BC)} > \Pi_B^{(0)}$ by symmetry of firm B and firm C) guarantees that small firms prefer staying in the small merger structure to operating as independent firms and, hence, prevents deviation (iii). Condition (2b) implies that, if one small firm (say, firm B), deviates from the small merger structure and merges with firm A, the total profit of firms A and B becomes smaller so that neither firm has an incentive to merge.⁹ Therefore, Condition (2b) ensures no small firm will deviate from the small merger structure to merge with firm A (i.e., deviation (iv) is prevented).

Figure 1 illustrates how Conditions (1) and (2) lead to a specific equilibrium structure. These two conditions are in fact not exclusive. Under the intersection of Conditions (1) and (2), there are two equilibrium structures: $\{AB, C\}$ and $\{A, BC\}$. To study firms' decisions on merger partner selection, we consider the independent structure $\{A, B, C\}$ as the status quo. That is, when two

merger structures $\{AB, C\}$ and $\{A, BC\}$ both fail to emerge as the equilibrium (i.e., neither Condition (1) nor Condition (2) holds), all firms prefer to stay independent, and the equilibrium structure is $\{A, B, C\}$. Note that, in this paper, we find that, after explicitly considering the equilibrium condition of $\{A, B, C\}$ based on the strong Nash equilibrium concept (i.e., $\Pi_A^{(0)} + \Pi_B^{(0)} \geq \Pi^{(AB)}$ and $\Pi_C^{(0)} \geq \Pi_C^{(BC)}$), the equilibrium market structure results are not changed.

4. When Small Firms Have Ample Capacity: $k \geq (1 - 2\alpha)$

To explicate our analysis, we now consider the case when all firms have ample capacity so that $k_A > k_B = k_C = k \geq (1 - 2\alpha)$, where $(1 - 2\alpha)$ is the sum of a firm's loyal buyers α and the disloyal buyers $(1 - 3\alpha)$. In this case, each small firm (B or C) has sufficient capacity k to satisfy the maximum potential demand (i.e., $1 - 2\alpha$) when the small firm offers the lowest price.

In the following, we first derive the profit of each firm under different structures and then identify the equilibrium structure by checking Conditions (1) and (2) through direct comparisons of profits across different structures.

4.1. Independent Structure $\{A, B, C\}$

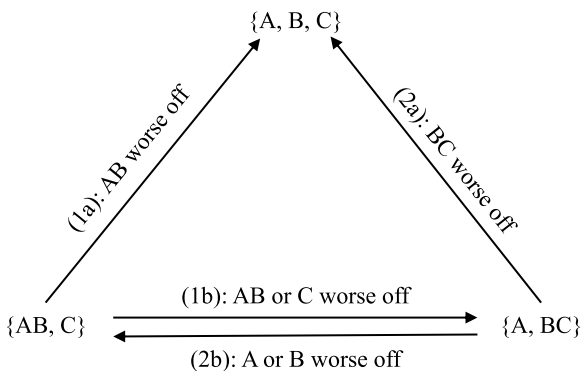
When $k \geq (1 - 2\alpha)$, the independent structure $\{A, B, C\}$ has three firms with ample capacity and a symmetric loyal segment. For each $i, j, h \in \{A, B, C\}$ such that $i \neq j \neq h$, firm i 's demand is

$$d_i(p_i, p_j, p_h) = \begin{cases} 0, & \text{if } p_i > r, \\ \alpha, & \text{if } \min\{p_j, p_h\} < p_i \leq r, \\ 1 - 2\alpha, & \text{if } p_i < \min\{p_j, p_h\} \text{ and } p_i \leq r, \\ \frac{1-\alpha}{2}, & \text{if } p_i = p_j < p_h, \text{ and } p_i \leq r, \\ \frac{1}{3}, & \text{if } p_i = p_j = p_h \leq r. \end{cases}$$

The demand from α loyal buyers is guaranteed for each firm, but each firm needs to balance between attracting disloyal switchers through price cutting and selling to fewer buyers by maintaining a higher price. Following standard arguments in the literature (Varian 1980, Narasimhan 1988), there is no pure-strategy equilibrium when $\alpha > 0$.¹⁰ When $\alpha = 0$, a pure-strategy equilibrium may exist.¹¹ Nevertheless, we find that, as α approaches zero, the expressions of firms' equilibrium profits and the equilibrium market structure for $\alpha > 0$ also converge to the results when $\alpha = 0$. For exposition brevity, we keep the most general statement that firms adopt the mixed pricing strategy in equilibrium.

Because three firms' demand functions are symmetric, we focus on the symmetric equilibrium in which each firm i sets its price p_i according to a certain cumulative

Figure 1. Effect of Conditions on Structure Transition



distribution function $F(\cdot)$ that can be shown to be defined over a range $[L, r]$, where L is the lower bound of price support (to be determined) and r is the buyer's reservation price (that is exogenously given). To determine $F(\cdot)$, L , and each firm's profit in equilibrium under structure $\{A, B, C\}$, first observe that, when each firm has ample capacity so that $k_A > k_B = k_C = k \geq (1 - 2\alpha)$, the firm that charges the lowest price can satisfy the demand from all switchers $(1 - 3\alpha)$ and its loyal buyers α , and each of the other two firms can just capture the demand of its α loyal buyers. Hence, when firm i sets its price p and firms j and h set their prices p_j and p_h , firm i 's expected profit $\pi_i(p)$ can be expressed as

$$\begin{aligned}\pi_i(p) &= \int_L^p \int_L^p p \alpha dF(p_j) dF(p_h) + \int_L^p \int_p^r p \alpha dF(p_j) dF(p_h) \\ &\quad + \int_p^r \int_L^p p \alpha dF(p_j) dF(p_h) \\ &\quad + \int_p^r \int_p^r p(1 - 2\alpha) dF(p_j) dF(p_h) \\ &= p(1 - 3\alpha)F^2(p) - 2p(1 - 3\alpha)F(p) + p(1 - 2\alpha),\end{aligned}$$

where $i, j, h \in \{A, B, C\}$ and $i \neq j \neq h$. By considering the first order condition associated with $\pi_i(p)$ and by using the fact that $F(L^{(0)}) = 0$ and $F(r) = 1$, we obtain the following result.

Lemma 1. *Consider the case when firms have ample capacity and operate in the independent structure $\{A, B, C\}$. In the symmetric equilibrium, each firm adopts the mixed strategy by setting the price according to a probability distribution $F(p)$ that is defined over the range $[L^{(0)}, r]$, where $L^{(0)} = \frac{r\alpha}{1-2\alpha}$. Also, all three firms' equilibrium profits are identical so that $\Pi_i^{(0)} = r\alpha$, $i = A, B, C$.*

The intuition behind Lemma 1 is as follows. First, because the guaranteed demand for each firm is α , each firm can obtain a guaranteed profit $r\alpha$ by setting its price to r . Second, notice that the maximum potential demand is $(1 - 2\alpha)$; hence, each firm's lowest acceptable price is $\frac{r\alpha}{1-2\alpha}$. By using the property of a mixed-strategy equilibrium, we can conclude that all three firms earn the same profit in equilibrium, where $\Pi_i^{(0)} = (1 - 2\alpha)L^{(0)} = r\alpha$. It is interesting to note that, as $\alpha \rightarrow 0$ (i.e., no loyal customers), each firm's profit goes to zero, which is consistent with the Bertrand equilibrium.

4.2. Merger Structure $\{ij, h\}$

When all three competing firms are symmetric in the sense that they all have ample capacity and identical loyal segments, it suffices to examine a generic two-firm merger structure $\{ij, h\}$ in which merged entity ij has 2α loyal buyers altogether. By using a similar approach as presented in Section 4.1, we derive the equilibrium results in Lemma 2.

Lemma 2. *Consider the case when firms have ample capacity and operate under a two-firm merger structure $\{ij, h\}$. In equilibrium, merged entity ij and firm h set their prices by following the cumulative distribution functions $F_{ij}(p)$ and $F_h(p)$, respectively, defined over the range $[L^{(ij)}, r]$, where $L^{(ij)} = \frac{2r\alpha}{1-\alpha}$. Also, the merged entity's profit $\Pi^{(ij)} = 2r\alpha$, and firm h 's profit $\Pi_h^{(ij)} = \frac{2r\alpha(1-2\alpha)}{1-\alpha}$.*

The intuition behind Lemma 2 is akin to that of Lemma 1. First, by setting the price to the buyer's reservation price r , merged entity ij and the stand-alone firm h can obtain their guaranteed profits of $2r\alpha$ and $r\alpha$, respectively. Second, observe from above that the highest potential demands for ij and the stand-alone firm h are $(1 - \alpha)$ and $(1 - 2\alpha)$, respectively. Hence, these two observations reveal that the lowest acceptable prices for the merged entity and the stand-alone firm are $\frac{2r\alpha}{1-\alpha}$ and $\frac{r\alpha}{1-2\alpha}$, respectively. In equilibrium, no entity sets a price strictly lower than its competitor with probability one so that the lower bound of price support $L^{(ij)} = \max\{\frac{2r\alpha}{1-\alpha}, \frac{r\alpha}{1-2\alpha}\} = \frac{2r\alpha}{1-\alpha}$.

Considering that firms have different lowest acceptable prices, we define the notion of pricing aggressiveness that will prove useful later as follows.

Definition 1. For an entity (a firm or a merged entity) in any structure, it is considered to be more pricing aggressive when its lowest acceptable price is smaller than that of the other entities.

To illustrate, it can be shown that the lowest acceptable price of merged entity ij (i.e., $\frac{2r\alpha}{1-\alpha}$) is higher than that of the stand-alone firm h (i.e., $\frac{r\alpha}{1-2\alpha}$), and thus, the stand-alone firm h is more pricing aggressive than merged entity ij in the merger structure $\{ij, h\}$.

4.3. Equilibrium Structure

We now derive the equilibrium structure by comparing the profits under different structures as stated in Lemmas 1 and 2 to examine whether Conditions (1) and (2) hold.

Proposition 1. *When firms have ample capacity so that $k_A > k_B = k_C = k \geq (1 - 2\alpha)$, the independent structure $\{A, B, C\}$ is the equilibrium structure.*

Observe that, under the independent structure, because all three firms have the same pricing aggressiveness, each firm earns its guaranteed profit, that is, $\Pi_A^{(0)} = \Pi_B^{(0)} = \Pi_C^{(0)} = r\alpha$. Under the merger structure $\{ij, h\}$, merged entity ij and stand-alone firm h both have ample capacity, but ij has more loyal buyers. Hence, merged entity ij is less pricing aggressive than h and obtains its guaranteed profit $2r\alpha$. As such Conditions (1a) and (2a) are violated. That is, no firm can benefit from deviating from the independent structure to be involved in a merger. Consequently, we can conclude

that, when all firms are not capacity constrained ($k \geq 1 - 2\alpha$), $\{A, B, C\}$ is the equilibrium structure.

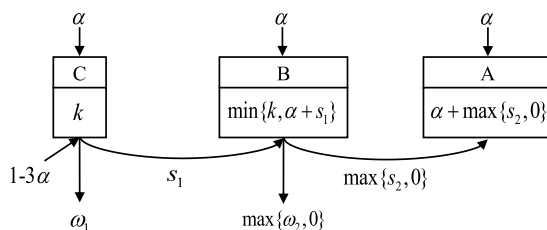
5. When Small Firms Have Limited Capacity: $\alpha < k < (1 - 2\alpha)$

We now investigate the three structures—*independent* $\{A, B, C\}$, *mixed merger* $\{AB, C\}$, and *small merger* $\{A, BC\}$ —for the case when firm A has ample capacity $k_A \geq (1 - 2\alpha)$ and yet small firms (B and C) have limited capacity $k \in (\alpha, 1 - 2\alpha)$. Unlike the ample capacity case as presented in Section 4, the analysis of this setting is much more complex because we need to allocate the spillover demand to other firms when the small firm cannot handle its own demand because of its limited capacity.

5.1. Independent Structure $\{A, B, C\}$

We first determine the demand and the realized sales for each firm by considering three different price scenarios. **Scenario 1:** Firm A offers the highest price. Without loss of generality, we assume that $p_A > p_B > p_C$. In this case, firm C's α loyal buyers and all switchers, in a total size of $(1 - 2\alpha)$, will first try to buy from firm C. Because of the capacity constraint, firm C's realized sales are k . By using the random rationing rule, each of these $(1 - 2\alpha)$ customers is served by firm C with probability $\frac{k}{1 - 2\alpha}$. Therefore, $\omega_1 \equiv \alpha(1 - \frac{k}{1 - 2\alpha})$ unsatisfied loyal buyers of firm C leave the market, and $s_1 \equiv (1 - 3\alpha)(1 - \frac{k}{1 - 2\alpha})$ unsatisfied switchers switch to firm B for purchase. Therefore, firm B's total demand is $(\alpha + s_1)$. As depicted in Figure 2, if firm B's capacity is still insufficient to meet its demand (i.e., $k < \alpha + s_1$), $\omega_2 \equiv \alpha(1 - \frac{k}{\alpha + s_1})$ unsatisfied loyal buyers leave the market, and $s_2 \equiv s_1(1 - \frac{k}{\alpha + s_1})$ unsatisfied switchers turn to firm A for purchase, and hence, the realized sales for firms A, B, and C are $(\alpha + s_2)$, k , and k , respectively. If $k \geq \alpha + s_1$, sales for firms A, B, and C are α , $(\alpha + s_1)$, and k . By using the same approach, we can derive each firms' sales in the other two scenarios. **Scenario 2:** Firm A offers the second lowest price (assume $p_B > p_A > p_C$). The realized sales for firms A, B, and C are $(\alpha + s_1)$, α , and k , respectively. **Scenario 3:** Firm A offers the lowest price. Firm A's sales are $(1 - 2\alpha)$, and firm B's (and C's) sales are α .

Figure 2. Rationing Process with Constrained Capacity When $p_A > p_B > p_C$



We can show that there is no pure-strategy equilibrium for such three-firm models because of the tension of selling to switchers and loyal buyers. The equilibrium outcomes are shown in Lemma 3 and are depicted in Figure 3.

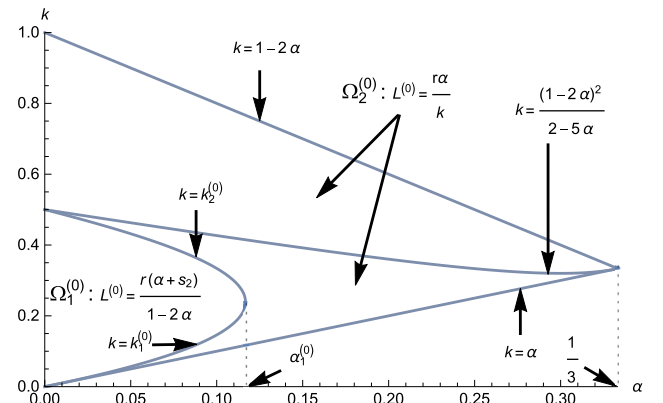
Lemma 3. Under the independent structure $\{A, B, C\}$, the lower bound of price support $L^{(0)}$ and the equilibrium profit of each firm satisfy¹²

- i. When $(\alpha, k) \in \Omega_1^{(0)}$, $L^{(0)} = \frac{r(\alpha + s_2)}{1 - 2\alpha}$, $\Pi_A^{(0)} = r(\alpha + s_2)$, and $\Pi_B^{(0)} = \Pi_C^{(0)} = \frac{r(\alpha + s_2)k}{1 - 2\alpha}$.
- ii. When $(\alpha, k) \in \Omega_2^{(0)}$, $L^{(0)} = \frac{r\alpha}{k}$, $\Pi_A^{(0)} = \frac{r\alpha(1 - 2\alpha)}{k}$, and $\Pi_B^{(0)} = \Pi_C^{(0)} = r\alpha$.

The above lemma can be interpreted as follows. First, consider region $\Omega_1^{(0)}$ (Figure 3) in which the size of loyal buyer α is small and yet $k > \alpha$ so that each firm has extra capacity to serve some disloyal buyers. In Lemma 3(i), $L^{(0)} = \frac{r(\alpha + s_2)}{1 - 2\alpha}$ is firm A's lowest acceptable price, which is greater than the small firms' lowest acceptable price $\frac{r\alpha}{k}$. Hence, small firms (B and C) are more pricing aggressive than firm A in region $\Omega_1^{(0)}$. This is mainly because small firms' guaranteed demand α is small but its capacity is not too low ($k > k_1^{(0)}$) so that it tries to compete fiercely for more demand in this region.

Next, in region $\Omega_2^{(0)}$ (Figure 3), firm A is more pricing aggressive, and $L^{(0)}$ is determined by the small firms' lowest acceptable price $\frac{r\alpha}{k}$. To understand it, we consider two zones. Zone 1 corresponds to $(1 - 2\alpha) > k \geq \frac{(1 - 2\alpha)^2}{2 - 5\alpha}$. In this case, small firms have large capacity to compete with firm A so that there are few unsatisfied switchers for firm A, and hence, firm A becomes more pricing aggressive than small firms. Zone 2 corresponds to $\frac{(1 - 2\alpha)^2}{2 - 5\alpha} > k > \alpha$. When k is close to $\frac{(1 - 2\alpha)^2}{2 - 5\alpha}$, the same reason in zone 1 leads firm A to be more pricing aggressive than small firms; when k is close to α , small firms are less aggressive because there is not much spare capacity to serve beyond their own loyal customers.

Figure 3. (Color online) Equilibrium Lower Bound of Price Support in Structure $\{A, B, C\}$ When $k \in (\alpha, 1 - 2\alpha)$



5.2. Small Merger Structure: $\{A, BC\}$

We now consider the small merger structure $\{A, BC\}$, in which merged entity BC has 2α loyal buyers and a capacity of $2k$ altogether, where $k < (1 - 2\alpha)$. We derive the equilibrium results in Lemma 4, and they are illustrated in Figure 4.

Lemma 4. Under the small merger structure $\{A, BC\}$, the lower bound of price support $L^{(BC)}$ and the equilibrium profit of each firm satisfy¹³

- When $(\alpha, k) \in \Omega_1^{(BC)}$, $L^{(BC)} = \frac{r(\alpha + s_3)}{1 - 2\alpha}$, $\Pi_A^{(BC)} = r(\alpha + s_3)$, and $\Pi^{(BC)} = \frac{2r(\alpha + s_3)k}{1 - 2\alpha}$, where $s_3 \equiv (1 - 3\alpha)(1 - \frac{2k}{1 - \alpha})$.
- When $(\alpha, k) \in \Omega_2^{(BC)}$, $L^{(BC)} = \frac{r\alpha}{k}$, $\Pi_A^{(BC)} = \frac{r\alpha(1 - 2\alpha)}{k}$, and $\Pi^{(BC)} = 2r\alpha$.
- When $(\alpha, k) \in \Omega_3^{(BC)}$, $L^{(BC)} = \frac{2r\alpha}{1 - \alpha}$, $\Pi_A^{(BC)} = \frac{2r\alpha(1 - 2\alpha)}{1 - \alpha}$, and $\Pi^{(BC)} = 2r\alpha$.

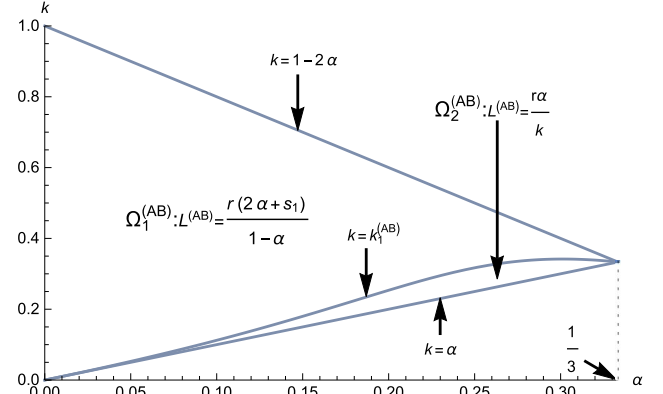
Figure 4 resembles Figure 3 as presented in Section 5.1 for the independent structure $\{A, B, C\}$. Consistent with our earlier result for the independent structure, Lemma 4 reveals that the small merged entity BC is more pricing aggressive than the large firm A in region $\Omega_1^{(BC)}$, where α is small and k is within an intermediate range. The underlying reason is similar to that under structure $\{A, B, C\}$ as discussed in Section 5.1, and hence, we omit the details.

5.3. Mixed Merger Structure: $\{AB, C\}$

Because firms B and C are symmetric, it suffices to examine the case in which the large firm A merges with a small firm (say, firm B). Merged entity AB operates like an integrated firm so that its capacity is unconstrained (because firm A has ample capacity) and its size of loyal buyers is 2α . Lemma 5, illustrated in Figure 5, provides the equilibrium outcomes in structure $\{AB, C\}$.

Lemma 5. Under the mixed merger structure $\{AB, C\}$, the lower bound of price support $L^{(AB)}$ and the equilibrium profit of each firm satisfy¹⁴

Figure 5. (Color online) Equilibrium Lower Bound of Price Support in Structure $\{AB, C\}$ When $k \in (\alpha, 1 - 2\alpha)$



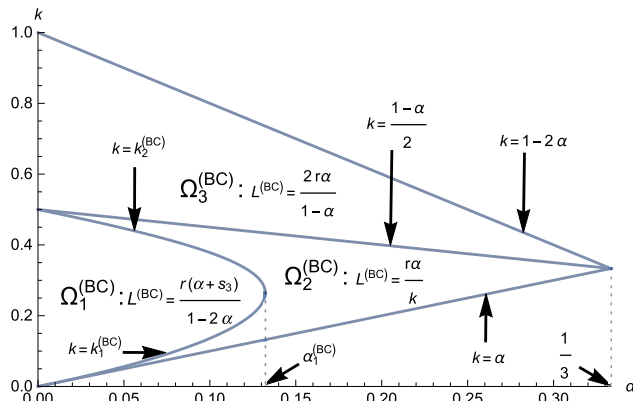
- When $(\alpha, k) \in \Omega_1^{(AB)}$, $L^{(AB)} = \frac{r(2\alpha + s_1)}{1 - \alpha}$, $\Pi^{(AB)} = r(2\alpha + s_1)$, and $\Pi_C^{(AB)} = \frac{r(2\alpha + s_1)k}{1 - \alpha}$.
- When $(\alpha, k) \in \Omega_2^{(AB)}$, $L^{(AB)} = \frac{r\alpha}{k}$, $\Pi^{(AB)} = \frac{r\alpha(1 - \alpha)}{k}$, and $\Pi_C^{(AB)} = r\alpha$.

From Lemma 5, it can be checked that, when the small firm's capacity k is sufficiently larger than α , which corresponds to a large region, the stand-alone firm C is more pricing aggressive than merged entity AB in region $\Omega_1^{(AB)}$. After the mixed merger, with an expanded base of loyal buyers, the merged entity AB is more concerned with the potential profit loss from loyal buyers when lowering prices and wants to maintain its profit margin without cutting its price too much. That is why the merged entity AB is less pricing aggressive than firm C in the majority of parameter settings ($\Omega_1^{(AB)}$) as illustrated in Figure 5. This is a negative driver for merging. Next, we proceed to analyze the effects of different types of mergers from the perspective of firms' capacity.

5.4. Asymmetric Capacity Effect

After a mixed merger, the previous large player's (firm A's) capacity is further expanded after merging with a small firm, say firm B. Then, the capacities of the resulting two competitors (merged entity AB and firm C) become more asymmetric than in the independent structure. This change in capacities is referred to as the asymmetric capacity effect of the mixed merger. This asymmetric capacity effect helps dampen market competition as follows. In the independent structure, firm A faces $2k$ competing capacity from two small firms. After a mixed merger between firms A and B, the competing capacity faced by the big player is reduced to k because of the capacity merging. Therefore, when the big player (firm A in $\{A, B, C\}$ and entity AB in $\{AB, C\}$) charges a higher price than small firms, the big player in the mixed merger structure can obtain more demand from switchers than that in the independent structure. As a result, the large entity and the stand-alone firm engage in less

Figure 4. (Color online) Equilibrium Lower Bound of Price Support in Structure $\{A, BC\}$ When $k \in (\alpha, 1 - 2\alpha)$



intense competition for switchers after the mixed merger, leading to a reduction in the overall market competition.

However, after a small merger, small firms' capacities are pooled together, leading to $2k$ units of capacity for the merged entity BC. In this case, the resulting two competitors' capacities become more symmetric compared with the independent case. Hence, the asymmetric capacity effect is absent in the small merger structure. Nevertheless, the small merger also may reduce market competition. The number of competitors in the market is reduced by one after two firms merge, resulting in a reduction in market competition. The competition-dampening effect of the small merger primarily comes from this aspect. The mixed merger can further dampen market competition by generating the asymmetric capacity effect.

Next, to support the above intuition, we investigate the impact of mergers on firms' selling prices, which can measure price competition intensity. Recall from our previous discussion that, under any given structure (independent, mixed merger, or small merger), typically there is a mixed-strategy equilibrium in which each entity sets its price based on a certain probability distribution $F(\cdot)$ with support $[L, r]$. A firm's lowest acceptable price is a measure of its pricing aggressiveness, but it may not be equal to the lower bound of its equilibrium price support. For instance, in the independent structure, if a firm's lowest acceptable price is lower than its competitors', in equilibrium it will not charge a price strictly below the competitors' lowest acceptable prices. Hence, when taking into account the pricing strategies of other firms, a firm's lowest acceptable price may be lower than the lower bound of its equilibrium price support. Note that, L is the lower bound of the equilibrium price support, which is determined by all firms' pricing strategies and can reflect the firms' pricing aggressiveness in the market. Therefore, one can interpret the lower bound of price support L as a measure of market competition associated with this structure. A structure involves softer market competition than another structure if its L is larger than another's.¹⁵ By using this interpretation of L , we now compare the degree of market competition across different structures by comparing $L^{(0)}$, $L^{(AB)}$, and $L^{(BC)}$ for different ranges of k in Online Table A1. This comparison leads to the following result.

Proposition 2. *The lower bounds of price support across the three structures for all ranges of k satisfy $L^{(AB)} \geq L^{(BC)} \geq L^{(0)}$.*

The result stated in Proposition 2 has two implications driven by the competition-dampening effect associated with different types of mergers. First, because $L^{(AB)} \geq L^{(BC)} \geq L^{(0)}$, we can conclude that mergers can enable firms to soften market competition. The intuition behind this is that the number of competitors is reduced after two firms merge. A similar result is found by Deneckere and Davidson (1985), who study a model in which firms

without capacity constraint produce symmetrically differentiated products. In our paper, firms have asymmetric capacities, and hence, we can further investigate how different types of mergers cause different scenarios of capacities and further different levels of market competition. This investigation results in the second implication of Proposition 2. Because $L^{(AB)} \geq L^{(BC)} \geq L^{(0)}$, we can conclude that market competition is the least intense in the mixed merger structure $\{AB, C\}$. The two competing parties in $\{AB, C\}$ are more asymmetric in capacity than the two parties in $\{A, BC\}$. The previously discussed asymmetric capacity effect of the mixed merger helps reduce the competition between merged entity AB and firm C, but this effect is absent for the small merger, leading to $L^{(AB)} \geq L^{(BC)}$.

Obviously, when the initial market competition is more fierce (i.e., higher small firms' capacity and lower number of loyal buyers), firms have greater incentives to mitigate market competition through mergers.

Besides the reduction in market competition caused by mergers, the decrease of unsatisfied loyal buyers of merging firms is another driver for merging. In the independent structure, when small firms' potential demand exceeds their capacity, some of their loyal buyers are not served and leave the market. After the mixed merger AB, all firm B's loyal buyers are satisfied because of firm A's unconstrained capacity. After the small merger BC, when the merged entity BC's capacity is unconstrained, the two small firms' loyal buyers can all be served; when BC's capacity is still constrained, some of their loyal buyers are rationed, but the chance of being served is increased because of the pooling of their capacities. That is, without considering changes in market prices, both the mixed merger and the small merger can effectively decrease the number of merging firms' unsatisfied loyal buyers and, hence, may increase merging firms' joint profit.

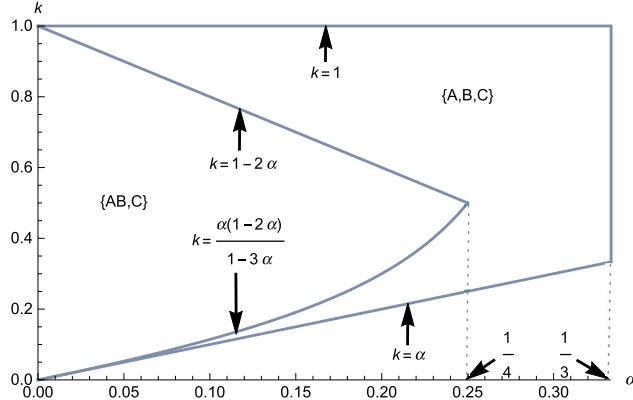
6. Equilibrium Merger Structure

To identify the equilibrium structure, we use the profits under different structures provided in Online Table A1 to examine whether Conditions (1) and (2) hold.¹⁶ We depict the equilibrium structure in Figure 6 and state our result as follows.

Proposition 3 (Equilibrium Structure). *For three firms with asymmetric capacity and symmetric size of loyal buyers, the mixed merger $\{AB, C\}$ is the equilibrium structure when the small firm's capacity $k \in \left(\frac{\alpha(1-2\alpha)}{1-3\alpha}, 1-2\alpha\right)$. Otherwise, the independent structure $\{A, B, C\}$ is the equilibrium.*

In the absence of a capacity constraint (i.e., when $k \geq 1-2\alpha$), the firms all have ample capacity. When two firms merge, the merged entity is less aggressive in price cutting than the stand-alone firm because it has more loyal buyers and wants to maintain its profit margin without cutting its price too much. Therefore, the

Figure 6. (Color online) A Full Picture of Equilibrium Structure



merged entity earns its guaranteed profit from loyal buyers, which equals the two participating firms' total profit in the independent structure. Thus, merging does not bring additional benefit, and the equilibrium structure is the independent one.

In the presence of a capacity constraint (i.e., when $k < 1 - 2\alpha$), first consider the special case when α approaches zero, that is, there are no loyal buyers. Observe from Proposition 3 that the condition $k \in \left(\frac{\alpha(1-2\alpha)}{1-3\alpha}, 1 - 2\alpha\right)$ becomes $k \in (0, 1)$, which always holds. That is, the mixed merger is always the equilibrium regardless of capacity levels as also illustrated in Figure 6. This implies that, when all buyers are price sensitive and, hence, competition is intense, the mixed merger is the equilibrium because of its most evident asymmetric capacity effect.

When $\alpha < 1/4$ and $k \in \left(\frac{\alpha(1-2\alpha)}{1-3\alpha}, 1 - 2\alpha\right)$, the size of loyal buyers is small, and yet the small firm's capacity k is reasonably large. In this case, both small firms need to compete for disloyal switchers to improve their earnings, and hence, the initial competition intensity is high. Then the same logic as above that, instead of staying independent to face fierce competition, the merger between the large firm A and a small firm B can soften competition the most so that all entities can charge higher prices as stated in Proposition 2; that is, the asymmetric capacity effect of a mixed merger is sufficiently strong so that structure $\{AB, C\}$ is the equilibrium in this region.

When $\alpha < 1/4$ and $k \in \left(\alpha, \frac{\alpha(1-2\alpha)}{1-3\alpha}\right)$, the capacity is not much larger than the size of loyal buyers (k close to α). As depicted in Figure 6, the mixed merger structure $\{AB, C\}$ is no longer an equilibrium, and instead, the independent structure $\{A, B, C\}$ becomes the equilibrium. The result is because, in this case, even though merging with a rival can enable firms to increase their price as stated in Proposition 2, this competition-dampening effect is not significant. Specifically, when k

is close to α , the small firms lack extra capacity to serve the switchers so that they do not compete fiercely with the large firm. Consequently, the initial competition intensity is low, and firms' incentive to reduce the market competition by merging is weak. Furthermore, after the mixed merger, the loyal buyers of the merged entity are expanded, and merging firms show greater concern for profit loss from loyal buyers by lowering prices. Hence, the likelihood for AB to charge higher prices than firm C is increased, in which case AB captures a lower demand from switchers. When k is close to α , this effect is dominant, and the merged entity AB fails to benefit from the positive asymmetric capacity effect and may even suffer a decline in profit. Hence, firms prefer to keep independent so that the independent structure $\{A, B, C\}$ is the equilibrium.

It remains to consider the region in which $\alpha > 1/4$. In this region, the condition $k \in \left(\frac{\alpha(1-2\alpha)}{1-3\alpha}, 1 - 2\alpha\right)$ in Proposition 3 is violated, and this observation implies that the independent structure $\{A, B, C\}$ is the equilibrium structure. Note that, when $\alpha > 1/4$, the small firms have a large size of loyal buyers, and hence, their guaranteed demand is sufficiently high so that they do not compete fiercely with the large firm. Then, the same logic applies here: the competition-dampening effect of mergers is weak, and the negative effect of the merging firms' reduced pricing aggressiveness dominates. Therefore, the independent structure $\{A, B, C\}$ is the equilibrium.

Unlike the random rationing rule in the main paper, we also explore a model with the efficient rationing rule under which firms' limited capacity is first allocated to loyal buyers and then to switchers (please refer to Online Appendix C for details). Different from the above results, the mixed merger structure $\{AB, C\}$ is the equilibrium when k is close to α under the efficient rationing rule. The reason is as follows. Under efficient rationing, when merged entity AB charges a higher price than firm C, the small firm C satisfies a smaller number of switchers, allowing entity AB to capture more switchers compared with the case under random rationing. Therefore, whereas the mixed merger increases the likelihood that merging firms charge higher prices than firm C under both rationing rules, this negative impact of decreased pricing aggressiveness is less significant under the efficient rationing rule. Consequently, under efficient rationing, even when k is close to α , the negative effect of a mixed merger cannot dominate its positive asymmetric capacity effect, and hence, firms A and B can benefit from merging (i.e., $\Pi^{(AB)} > \Pi_A^{(0)} + \Pi_B^{(0)}$).

Next, we explain why the small merger structure $\{A, B, C\}$ does not emerge as the equilibrium structure. Similar to the mixed merger, the small merger BC dampens the market competition by reducing the number of competing firms and decreases the number of merging

firms' unsatisfied loyal buyers. However, different from the mixed merger, the small merger BC cannot strengthen capacity asymmetry in the market so that there is no asymmetric capacity effect. Hence, the small merger's competition-dampening effect is weaker than that of the mixed merger. In addition, the small merger can also generate a negative effect because of the reduction in merging firms' pricing aggressiveness. Therefore, in the absence of the positive asymmetric capacity effect and the presence of the negative effect for the small merger, the small merger either does not improve participating firms' profit (in all regions other than $\Omega_1^{(BC)}$ in Figure 4) or it can improve participating firms' profit but firms A and B have incentives to merge by jointly deviating from the small merger structure (in $\Omega_1^{(BC)}$). Consequently, the small merger cannot be sustained as an equilibrium structure.

In summary, when all three firms have the same size of loyal buyers, only the independent structure $\{A, B, C\}$ and the mixed merger structure $\{AB, C\}$ are possible to emerge as the equilibrium, and the small merger structure $\{A, BC\}$ will never be sustainable.¹⁷ Will the small merger structure ever emerge as the equilibrium when firms have different loyal segment sizes? We investigate this question in Section 8.

7. Social Welfare and Consumer Surplus

In this section, we investigate the impacts of different mergers on social welfare and consumer surplus. Conventional wisdom suggests mergers that can generate impressive synergies or economies of scale may improve consumer surplus and/or social welfare (Perry and Porter 1985, Farrell and Shapiro 1990). Nevertheless, synergies and efficiency improvement are easy to promise yet often hard to deliver (Farrell and Shapiro 1990). Our paper provides an opportunity to examine whether mergers without synergies or economies of scale always reduce consumer surplus and social welfare. We also investigate how firms' capacities and consumer structure mediate the impacts of horizontal mergers on these two measures. Because the expressions of social welfare and consumer surplus are too complex in region $\{(\alpha, k) | 0 < \alpha < \frac{1}{3}, \frac{(1-2\alpha)^2}{2-5\alpha} < k < 1-2\alpha\}$ (i.e., the upper region of $\Omega_2^{(0)}$ in Figure 3), theoretical analysis is not amenable, and we resort to a numerical approach. However, we can analytically analyze the impact of mergers in the other region $\{(\alpha, k) | 0 < \alpha < \frac{1}{3}, \alpha < k < \frac{(1-2\alpha)^2}{2-5\alpha} \text{ or } 1-2\alpha < k < 1\}$. We summarize the analytical comparison results in Proposition 4, and based on these results, we provide managerial implications from government policy perspective. All theoretical and numerical results are illustrated in Figure 7.

Proposition 4 (Comparison Results of Social Welfare and Consumer Surplus Among Three Market Structures). *In*

region $\{(\alpha, k) | 0 < \alpha < \frac{1}{3}, \alpha < k < \frac{(1-2\alpha)^2}{2-5\alpha} \text{ or } 1-2\alpha < k < 1\}$, we have¹⁸

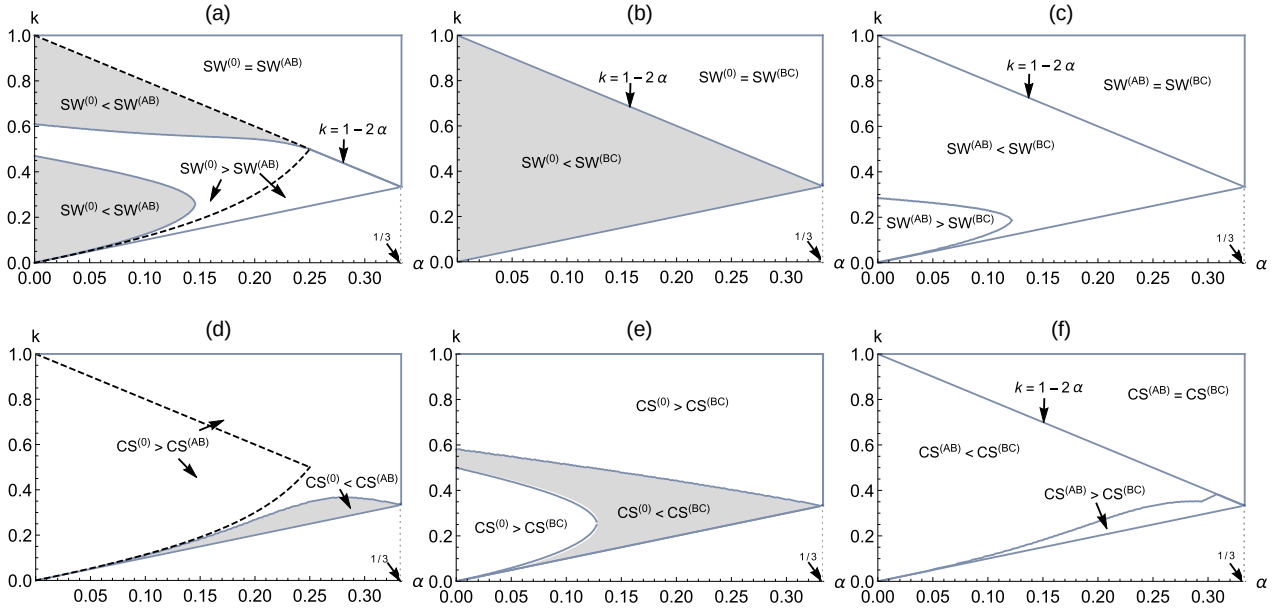
- i. If $(0 < \alpha < \alpha_1^{SW} \text{ and } k_2^{SW} < k < k_1^{SW})$ or $(\alpha_1^{SW} < \alpha < \alpha_2^{SW} \text{ and } k_2^{SW} < k < k_3^{SW})$, $SW^{(0)} < SW^{(AB)}$; if $0 < \alpha < 1/3$ and $1-2\alpha < k < 1$, $SW^{(0)} = SW^{(AB)}$; otherwise (exclusive of boundaries, the same hereafter), $SW^{(0)} > SW^{(AB)}$.
- ii. If $0 < \alpha < 1/3$ and $\alpha < k < \frac{(1-2\alpha)^2}{2-5\alpha}$, $SW^{(0)} < SW^{(BC)}$; otherwise, $SW^{(0)} = SW^{(BC)}$.
- iii. If $(0 < \alpha < \alpha_4^{SW} \text{ and } k_6^{SW} < k < k_5^{SW})$ or $(\alpha_4^{SW} < \alpha < \alpha_3^{SW} \text{ and } k_4^{SW} < k < k_5^{SW})$, $SW^{(AB)} > SW^{(BC)}$; if $0 < \alpha < 1/3$ and $1-2\alpha < k < 1$, $SW^{(AB)} = SW^{(BC)}$; otherwise, $SW^{(AB)} < SW^{(BC)}$.
- iv. If $0 < \alpha < 1/3$ and $\alpha < k < \min\{k_2^{CS}, \frac{(1-2\alpha)^2}{2-5\alpha}\}$, $CS^{(0)} < CS^{(AB)}$; otherwise, $CS^{(0)} > CS^{(AB)}$.
- v. If $(0 < \alpha < \alpha_2^{CS} \text{ and } k_3^{CS} < k < k_4^{CS})$ or $(0 < \alpha < 1/3 \text{ and } 1-2\alpha < k < 1)$, $CS^{(0)} > CS^{(BC)}$; otherwise, $CS^{(0)} < CS^{(BC)}$.
- vi. If $0 < \alpha < 1/3$ and $\alpha < k < \min\{k_5^{CS}, \frac{(1-2\alpha)^2}{2-5\alpha}\}$, $CS^{(AB)} > CS^{(BC)}$; If $0 < \alpha < 1/3$ and $1-2\alpha < k < 1$, $CS^{(AB)} = CS^{(BC)}$; otherwise, $CS^{(AB)} < CS^{(BC)}$.

As shown in Proposition 4, different from the literature, we find that mergers without synergies or economies of scale, although they dampen market competition and raise selling prices, can increase both social welfare and consumer surplus.¹⁹ The reason is that, in the presence of firms' capacity constraints, a merger can mitigate the participating small firm's capacity constraint, and hence, fewer buyers are not satisfied. Furthermore, according to Proposition 4, neither a mixed merger nor small merger can improve social welfare or consumer surplus when all firms have ample capacities (i.e., $k > 1-2\alpha$). This suggests that the presence of a capacity constraint is essential for mergers to increase social welfare and consumer surplus.

According to equilibrium structure results (Proposition 3) and Proposition 4(i), when $\{AB, C\}$ is the equilibrium structure, the AB merger may be socially beneficial. Therefore, the first policy implication of our result is as follows: a merger between a large firm and a small firm may be proposed in practice; the antitrust agencies can approve this merger to improve social welfare if small firms have relatively low capacity and few loyal buyers or small firms' capacity is sufficiently high but limited (please see Figure 7(a) for details). However, Figure 7(d) illustrates that the AB merger always decreases consumer surplus when $\{AB, C\}$ is the equilibrium. Hence, if the agency pays more attention to consumer surplus, then all proposed mergers should not be permitted.

Combining Propositions 3 and 4, we find that, depending on the parameter settings, each merger structure can result in the highest social welfare or consumer surplus but may not be the equilibrium market structure. Therefore, the government can appropriately promote mergers

Figure 7. (Color online) Comparison Results of Social Welfare and Consumer Surplus Among Three Market Structures with Symmetric Loyal Buyers



Notes. In panels (a) and (d), in the region enclosed by the dashed lines/curves and the vertical axis, $\{A, B, C\}$ is the equilibrium (see Figure 6). (a) $SW^{(0)}$ vs. $SW^{(AB)}$. (b) $SW^{(0)}$ vs. $SW^{(BC)}$. (c) $SW^{(AB)}$ vs. $SW^{(BC)}$. (d) $CS^{(0)}$ vs. $CS^{(AB)}$. (e) $CS^{(0)}$ vs. $CS^{(BC)}$. (f) $CS^{(AB)}$ vs. $CS^{(BC)}$.

that are beneficial for society and/or consumers by adopting some restrictive measures or offering incentives. This is the second policy implication. For example, Proposition 4, (ii) and (v), implies that merger between two small firms can increase social welfare and/or consumer surplus under certain parameter settings. Furthermore, small firms do have an incentive to merge (i.e., $\Pi_C^{(BC)} > \Pi_C^{(0)}$) if α is small but $\{A, BC\}$ is not an equilibrium because it is dominated by $\{AB, C\}$ (please see Online Figure A1 and Online Table A2). Therefore, in this case, the government can explicitly prohibit mergers with a large firm (for example, in the United States, according to the U.S. Department of Justice and Federal Trade Commission (2010), the commission is likely to challenge mergers that would significantly increase concentration), and then the merger between small firms would be proposed. Our result suggests that the government can approve this merger in such cases because this merger can mitigate stockout risk for small firms but just slightly or will not dampen market competition.

8. Asymmetric Loyal Segments

In this section, we extend our analysis to the case when the large firm A has more loyal buyers than each small firm. That is, each small firm has the same size of loyal segment α , and yet the large firm A has a larger loyal segment of size $l\alpha$, where $l > 1$.²⁰ As before, to ensure that the size of all loyal segments (i.e., $(2 + l)\alpha$) does not exceed the size of the buyer market (i.e., 1), $\alpha < \frac{1}{2+l}$.

8.1. Profits and Lower Bounds of Price Support in Different Structures

By using the same approach as presented in Sections 4 and 5, we can determine equilibrium profits and lower bounds of price supports in different structures. Interestingly, we find that the three firms' equilibrium price supports may differ in the independent structure when firm A has a larger loyal segment and the small firms' capacity level is above a threshold as shown in the following proposition.

Proposition 5 (Equilibrium Lower Bounds of Price Support). *When the firms have asymmetric loyal buyers,²¹*

i. *In the independent structure $\{A, B, C\}$, when $k < \bar{k}$, the three firms set prices over an identical support with the lower bound $L^{(0)} = \max\left\{\frac{r\alpha}{k}, \frac{r(l\alpha + \max\{\bar{s}_2, 0\})}{1-2\alpha}\right\}$; when $k \geq \bar{k}$, firm A sets price at r with probability one and the two small firms compete with each other by setting prices below r with the lower bound of price support $L^{(0)} = \frac{r(\alpha + \bar{s}_1)}{k}$ for $k < 1 - \alpha - l\alpha$ and $L^{(0)} = \frac{r\alpha}{1-\alpha-l\alpha}$ for $k \geq 1 - \alpha - l\alpha$.*

ii. *In the mixed merger structure $\{AB, C\}$, merged entity AB and firm C set prices over an identical support with the lower bound $L^{(AB)} = \max\left\{\frac{r\alpha}{k}, \frac{r(l\alpha + l\alpha + \bar{s}_1)}{1-\alpha}\right\}$ for $k < 1 - \alpha - l\alpha$ and $L^{(AB)} = \frac{r(\alpha + l\alpha)}{1-\alpha}$ for $k \geq 1 - \alpha - l\alpha$.*

iii. *In the small merger structure $\{A, BC\}$, firm A and merged entity BC set prices over an identical support with the lower bound $L^{(BC)} = \max\left\{\frac{2r\alpha}{\min\{2k, 1-l\alpha\}}, \frac{r(l\alpha + \max\{\bar{s}_3, 0\})}{1-2\alpha}\right\}$ for $k < 1 - \alpha - l\alpha$ and $L^{(BC)} = \max\left\{\frac{2r\alpha}{1-l\alpha}, \frac{r\alpha}{1-2\alpha}\right\}$ for $k \geq 1 - \alpha - l\alpha$.*

In $\{A, B, C\}$, because the large firm A has more loyal buyers than the two small firms, firm A needs to decide whether to compete with the small firms for switchers. When the small firms' capacity is high enough $k \geq \bar{k}$, the small firms are relatively aggressive in pricing, and firm A is better off not competing with the small firms and simply setting its price at the reservation price r for certainty. Therefore, for independent structure, the equilibrium results when $k \geq \bar{k}$ are interpreted as follows: $L^{(0)}$ is equal to the lower bound of the small firms' price support but not equal to firm A's as firm A never prices below r . However, we can still use $L^{(0)}$ as a measure of market competition.

8.2. Equilibrium Merger Structure

Recall from Proposition 2 that mergers can dampen price competition to different levels. The following proposition examines whether this result persists when the loyal segment size is firm specific.

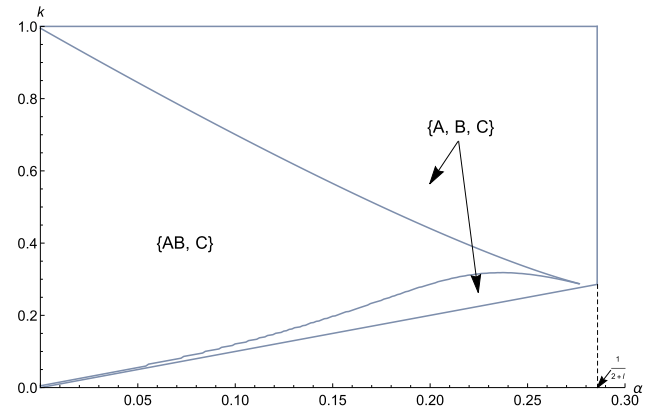
Proposition 6. *When firms have asymmetric loyal buyers, the equilibrium lower bounds of price support across the three structures satisfy $L^{(AB)} \geq L^{(BC)} \geq L^{(0)}$.*

Analogous to Proposition 2, Proposition 6 reveals that the competition-dampening effect of mergers persists in the case of asymmetric loyal buyers. Furthermore, this effect is more pronounced under the mixed merger because of an extra enhanced asymmetric capacity effect.

Next, we examine how the ratio of the large firm's loyal buyer size over small firms' l influences the equilibrium merger structure. By considering the equilibrium Conditions (1) and (2), we can derive the equilibrium merger structure for any given values of $l(> 1)$. We find that there is a threshold $l^*(= 2)$ such that, when $l \leq l^*$, the equilibrium outcome is similar to the result of the base model with $l = 1$. Actually, the equilibrium outcome in the extension model converges to the result of the base case when $l \rightarrow 1$. For example, consider the equilibrium merger structure as shown in Figure 8 for the case of $l = 1.5$. Akin to the equilibrium merger structure as shown in Figure 6 in the base model, we observe from Figure 8 that the mixed merger $\{AB, C\}$ emerges as the equilibrium structure when k is medium and α is not too large, and the range of k that corresponds to the mixed merger equilibrium shrinks with a larger α . In other regions, the independent structure $\{A, B, C\}$ is the equilibrium and the small merger $\{A, BC\}$ does not emerge as the equilibrium structure.

When considering the case in which $l > l^*$ as illustrated in Figure 9, we find that the mixed merger structure $\{AB, C\}$ is the equilibrium for small α and intermediate k as before. Interestingly however, the small merger structure $\{A, BC\}$ can also be the equilibrium when l is sufficiently large. For instance, as shown in Figure 9, the small merger structure $\{A, BC\}$ emerges

Figure 8. (Color online) Equilibrium Structure When $l = 1.5$



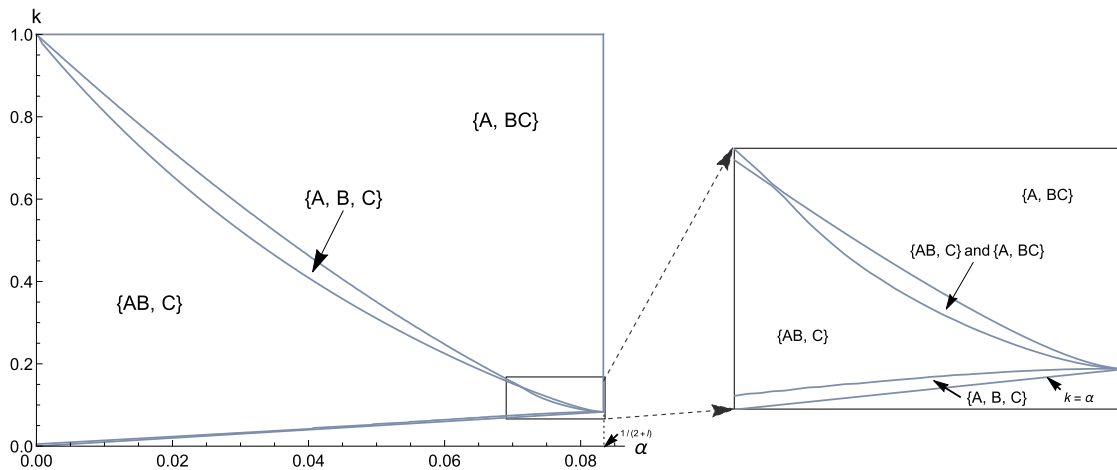
as the equilibrium when the small firm's capacity k is sufficiently large.

Note that, in the base model when $l = 1$, $\{A, BC\}$ cannot emerge as the equilibrium because of the absence of the positive asymmetric capacity effect. However, such results for sufficiently large k change in the asymmetric loyal segment setting because the asymmetric capacity effect of the small merger is no longer absent. We explain the logic as follows.

In the independent structure, when firm A has a much larger loyal segment than small firms, the small firms become so aggressive because they have few loyal buyers but large capacity k to capture disloyal switchers. In this case, being even more aggressive than small firms such as in the case of symmetric loyal buyers is no longer optimal for firm A. Instead, staying less aggressive than small firms by charging r with certainty is a better choice. That is, in structure $\{A, B, C\}$, facing two fierce competitors B and C, firm A with a large size of loyal buyers does not engage in active competition and just the two symmetric small firms compete intensely in price (please refer to Proposition 5(i) for details). This implies that the capacities of active competing firms are actually symmetric.

In contrast, in the small merger structure $\{A, BC\}$, the intensive competition between firms B and C is avoided. Most importantly, now the capacities of two competitors A and BC are asymmetric. That is, the active competitors' capacities switch from being symmetric in $\{A, B, C\}$ to asymmetric in $\{A, BC\}$ after the small merger when k is sufficiently large. In this case, the small merger can generate the positive asymmetric capacity effect, which can significantly reduce price competition. This is in sharp contrast with the result in the model with symmetric loyal segments that the small merger does not generate an asymmetric capacity effect. Furthermore, because firm A has a much larger base of loyal buyers, the likelihood that the merging firms charge higher prices than

Figure 9. (Color online) Equilibrium Structure When $l = 10$



the stand-alone firm is higher in structure $\{AB, C\}$ than in structure $\{A, BC\}$. That is, the negative effect of reduced pricing aggressiveness for merger AB is stronger. Therefore, although the mixed merger also generates an asymmetric capacity effect, it may be dominated by the small merger. The possible existence of the small merger's asymmetric capacity effect in the model with asymmetric loyal segments explains why the small merger can emerge as the equilibrium when both l and k are large as shown in Figure 9.

As a note, the regions for $\{A, BC\}$ and for $\{AB, C\}$ to be the equilibrium structure are not exclusive. As shown in the enlarged area in Figure 9, both merger structures are equilibria in the intersection of the corresponding two regions.

The comparison results of social welfare and consumer surplus among the three structures in this extension model with asymmetric loyal buyers are illustrated in Online Figure A2. Akin to the results in the base model (Proposition 4 and Figure 7), we find that, when $\{AB, C\}$ is the equilibrium market structure, the AB merger may be socially beneficial. Furthermore, we find that when k is sufficiently high but limited, $\{A, BC\}$ can emerge as the equilibrium, and meanwhile, the BC merger can increase both social welfare and consumer surplus. In this case, a proposed merger between two small firms should be approved.

9. Concluding Remarks

This paper studies voluntary merger decisions among a large firm (firm A) and two small firms (B and C) that compete in price in the market. We explore the effects of horizontal mergers by focusing on the characteristics of firms' capacities and loyal buyers. We identify that mergers can dampen the overall market competition by increasing firms' capacity asymmetry. This positive driver for merging is referred to as the asymmetric

capacity effect. Conversely, the concentration of loyal buyers can decrease merging firms' pricing aggressiveness versus the stand-alone firm, which reduces their demand from switchers under the random rationing rule. This is a negative driving force for merging.

In the base model in which all three firms have a symmetric size of the loyal segment, our analysis shows that only the mixed merger structure or the independent structure can be the equilibrium. The small merger structure cannot emerge as the equilibrium because the small merger cannot generate the positive asymmetric capacity effect.

Between the mixed merger structure $\{AB, C\}$ and the independent structure $\{A, B, C\}$, our results show that the former is the equilibrium outcome if all firms' loyal buyer size is sufficiently low and the small firms' capacity is sufficiently large but constrained (which implies relatively intense market competition), and otherwise, the independent structure is the equilibrium. This is because the mixed merger's asymmetric capacity effect, which significantly dampens competition is amplified for a highly competitive market. When the small firms' capacity is too low or the size of loyal segment is large, the initial market competition in the independent structure is not intense, in which case the positive asymmetric capacity effect is not significant and the negative effect of mergers dominates, and thus, all three firms stay independent. When the small firms' capacity is not constrained, the three firms all have ample capacity. In this case, although a merger between two firms can dampen price competition, a merger does not benefit the merging firms, and hence, in equilibrium, the firms stay independent.

When the large firm has a much larger loyal segment than the small firms, interestingly, the small merger structure $\{A, BC\}$ can emerge as the equilibrium. This is because the small merger can result in an asymmetric

capacity effect on some occasions as follows. When the small firms have a large capacity but a much smaller loyal segment, in the independent structure, the large firm opts for not competing with the small firms, but the small firms compete intensively with each other; that is, the capacities of active competitors are symmetric. In such a scenario, the merger between two small firms creates capacity asymmetry between the large firm and the merged entity by small firms, hence resulting in an evident positive asymmetric capacity effect. Therefore, the small merger structure $\{A, BC\}$ can emerge as the equilibrium.

Our finding that the mixed merger is more likely to emerge as the equilibrium is consistent with the observation in practice: mergers involving large firms (e.g., AB) are more common than small mergers (e.g., BC). In reality, some merged firms may cut their production capacity. However, they should not have incentives to decrease capacity to a level below its maximum potential demand because this behavior can reduce the merged entity's pricing aggressiveness and benefit the stand-alone firm but not the merged entity.

This paper also studies the impacts of mergers on social welfare and consumer surplus to deliver implications from the antitrust perspective. In contrast to a large literature on mergers, we find that capacity merging without synergies or economies of scale may increase social welfare and consumer surplus though mergers can dampen market competition and raise selling prices. This is because mergers can mitigate stockout risk for small firms whose capacity is constrained. More specifically, a mixed merger, when emerging as the equilibrium, can improve social welfare but decrease consumer surplus. Hence, the antitrust agencies can approve a mixed merger to improve social welfare under certain conditions, but if the agencies pay more attention to consumer surplus, then mixed mergers should not be permitted. Furthermore, because small mergers can improve social welfare and/or consumer surplus, the government can appropriately promote such beneficial mergers by adopting some restrictive measures or offering incentives.

We focus on a three-firm case in our model, and the large firm's capacity is unconstrained. We analyze a four-firm model with three levels of capacity and a four-firm model with three levels of loyal segment in Online Appendix G to show the robustness of our results. We also study a model with the large firm's capacity constrained in Online Appendix F. The corresponding results confirm and enrich our understanding of the underlying driving forces.

This research can be further extended in several directions. First, it will be promising to investigate the role of merger operation mode in merger decisions. Our paper considers that, once a merger occurs, the two merging

firms act as a single firm by pooling together their capacities and loyal buyers, which can be viewed as a complete merger mode. Except this, there are other merger modes as follows: Before the merger, firms can contract on the terms of merger profit allocation, and thus, each merging firm still tries to maximize its own profit. In a market with multiple competing firms, several or all firms can merge just to share their capacities to capture the unsatisfied buyers, but each firm keeps its loyal buyers. Second, it remains to study horizontal mergers in other market structures, for example, two large firms and one small firm or three firms different from each other.

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Endnotes

¹ A semiconductor company that designs, manufactures, and sells integrated circuits is an IDM, and it competes with foundries in semiconductor production.

² Yin (2010) also adopts the strong Nash equilibrium concept.

³ When there are multiple firms offering the same lowest price, we assume these switchers buy from each of these firms with equal probability.

⁴ If $k \leq \alpha$, then the sales of both firms B and C are k as long as $p_i \leq r, i = B, C$, and the problem is trite.

⁵ To keep our focus on the equilibrium merger structure, we do not impose any specific rule to allocate the profit between firms A and B throughout this paper.

⁶ The presence of the fixed transition fee determines whether inequalities in Condition (1) are strict or nonstrict. To be specific, denote the arbitrarily small transition fee as $\epsilon > 0$. In $\{AB, C\}$, the net profit of the merged entity AB is $\Pi^{(AB)} - \epsilon$. In $\{A, BC\}$, because firms B and C are symmetric, each small firm bears half of the merging cost, $\epsilon/2$, and hence, each small firm's net profit is $\Pi_i^{(BC)} - \epsilon/2$, $i = B, C$. For example, conditions $\Pi^{(AB)} - \epsilon \geq \Pi_A^{(0)} + \Pi_B^{(0)}$ and $\Pi_C^{(AB)} \geq \Pi_C^{(BC)} - \epsilon/2$ prevent deviations (i) and (ii), respectively. As ϵ is arbitrarily small but positive, the first inequality is equivalent to $\Pi^{(AB)} > \Pi_A^{(0)} + \Pi_B^{(0)}$, whereas the second is equivalent to $\Pi_C^{(AB)} \geq \Pi_C^{(BC)}$. Note that, the condition $\Pi^{(AB)} = \Pi_A^{(0)} + \Pi_B^{(0)}$ cannot prevent deviation (i) because of the existence of the positive transition fee, that is, $\Pi_A^{(0)} + \Pi_B^{(0)} = \Pi^{(AB)} > \Pi^{(AB)} - \epsilon$. We provide the detailed derivation of equilibrium Conditions (1) and (2) that involve an arbitrarily small transition fee in Online Appendix B.

⁷ This condition is general, and it does not rely on a specific rule to allocate the profit between firms A and B. By focusing on the total profit $\Pi^{(AB)}$, we know for sure that, if Condition (1a) holds, it is possible for both firms A and B to obtain a larger profit in the mixed merger than in the independent structure $\{A, B, C\}$, that is, it is possible to split the total profit $\Pi^{(AB)}$ so that $\Pi_A^{(AB)} > \Pi_A^{(0)}$ and $\Pi_B^{(AB)} > \Pi_B^{(0)}$; otherwise, it is impossible for firms A and B to merge.

⁸ Firm A has no incentive to switch its merger partner from firm B to firm C because firms B and C are symmetric.

⁹ When switching from $\{A, BC\}$ to $\{AB, C\}$, either firm A's profit or firm B's profit is decreased. If firm A's profit is decreased, firm A will not accept firm B's invitation to merge with it; if firm B's profit becomes smaller, firm B will not deviate either.

¹⁰ The intuition behind this is as follows. Suppose three firms' prices satisfy $p_i \geq p_j \geq p_h$. If $p_j > p_h$, then the firm that charges p_h can always increase its profit by raising its price arbitrarily close to but still lower than p_j . If $p_j = p_h \neq 0$, then one of these two firms can undercut price by a sufficiently small amount to attract all the switchers and get better off. If $p_j = p_h = 0$, one of these two firms can deviate by raising its price to r and obtain positive profits from loyal buyers. Hence, there does not exist a pure-strategy pricing equilibrium.

¹¹ Specifically, in the absence of loyal buyers, if $k \geq \frac{1}{2}$, there exists a pure-strategy equilibrium under $\{A, B, C\}$ and $\{A, BC\}$, and if $k \geq 1$, a pure-strategy equilibrium exists under $\{AB, C\}$; otherwise, there is no pure-strategy equilibrium. However, in the absence of a capacity constraint, the pure-strategy equilibrium always exists in our price-setting game when $\alpha = 0$ and firms' production costs are strictly convex (Dastidar 1995).

¹² Here, the regions are defined as $\Omega_1^{(0)} = \{(\alpha, k) | 0 < \alpha < \alpha_1^{(0)}, k_1^{(0)} < k < k_2^{(0)}\}$, $\Omega_2^{(0)} = \{(\alpha, k) | (\alpha, k) \notin \Omega_1^{(0)}, 0 < \alpha < 1/3, \alpha < k < 1 - 2\alpha\}$, where $k_1^{(0)}(\alpha) = \frac{(1-2\alpha)[(1-\alpha)(1-3\alpha) - \sqrt{(1-3\alpha)(-43\alpha^3 + 43\alpha^2 - 13\alpha + 1)}]}{2(1-3\alpha)(2-5\alpha)}$, $k_2^{(0)}(\alpha) = \frac{(1-2\alpha)[(1-\alpha)(1-3\alpha) + \sqrt{(1-3\alpha)(-43\alpha^3 + 43\alpha^2 - 13\alpha + 1)}]}{2(1-3\alpha)(2-5\alpha)}$, and $k_1^{(0)}(\alpha_1^{(0)}) = k_2^{(0)}(\alpha_1^{(0)})$.

¹³ Here, the regions are defined as $\Omega_1^{(BC)} = \{(\alpha, k) | 0 < \alpha < \alpha_1^{(BC)}, k_1^{(BC)} < k < k_2^{(BC)}\}$, $\Omega_2^{(BC)} = \{(\alpha, k) | (\alpha, k) \notin \Omega_1^{(BC)}, 0 < \alpha < 1/3, \alpha < k < \frac{1-\alpha}{2}\}$, $\Omega_3^{(BC)} = \{(\alpha, k) | 0 < \alpha < 1/3, \frac{1-\alpha}{2} \leq k < 1 - 2\alpha\}$, where $k_1^{(BC)} = \frac{(1-\alpha)(1-2\alpha) - \sqrt{(1-\alpha)(1-2\alpha)(26\alpha^2 - 11\alpha + 1)}}{4(1-3\alpha)}$, $k_2^{(BC)} = \frac{(1-\alpha)(1-2\alpha) + \sqrt{(1-\alpha)(1-2\alpha)(26\alpha^2 - 11\alpha + 1)}}{4(1-3\alpha)}$, and $k_1^{(BC)}(\alpha_1^{(BC)}) = k_2^{(BC)}(\alpha_1^{(BC)})$.

¹⁴ Here, the two regions are defined as $\Omega_1^{(AB)} = \{(\alpha, k) | 0 < \alpha < 1/3, k_1^{(AB)} \leq k < 1 - 2\alpha\}$, $\Omega_2^{(AB)} = \{(\alpha, k) | 0 < \alpha < 1/3, \alpha < k < k_1^{(AB)}\}$, where $k_1^{(AB)}(\alpha) = \frac{(1-\alpha)(1-2\alpha) - \sqrt{(1-\alpha)(1-2\alpha)(14\alpha^2 - 7\alpha + 1)}}{2(1-3\alpha)}$.

¹⁵ In a mixed-strategy equilibrium, another measure of a firm's pricing aggressiveness is the firm's expected price. However, we find that measuring the change in market competition intensity by comparing firms' expected prices before and after a merger is unsuitable for our modeling framework for the following reason. The effects of a merger on the merged entity's and the stand-alone firm's expected prices may be opposite. When one entity's expected price increases but the other's decreases, it is hard to tell whether the competition is intensified or dampened. Therefore, we cannot use the expected price of one party to measure the price competition degree of the whole market.

¹⁶ These comparisons result in eight regions as shown in Online Figure A1. In Online Table A2, we summarize for each region, the profit comparison, whether Conditions (1) and (2) hold, and the equilibrium structure.

¹⁷ To show the robustness of our findings in different model frameworks, we analytically study two models in which firms produce differentiated products: one model in which firms compete in quantity and the other in which firms compete in price. We consider both cases with and without capacity constraint in the two models. Different from our model, customer loyalty is not considered in such a model framework, and these models cannot be easily extended to the case with asymmetric demand structures in studying horizontal mergers. In the quantity competition model, the result shows that, similar to our base model, only $\{AB, C\}$ and $\{A, B, C\}$ can emerge as the equilibrium. In the price competition model, similar to our base model, $\{AB, C\}$ is the equilibrium when the initial market competition is intense (i.e., α is small in our model), and the degree of product substitution is high in the new model; under structure $\{A, BC\}$, the small merger also can increase small firms' profit for certain parameter settings, but the stand-alone firm A is willing to take a free ride and has no incentive to merge with a small firm, and hence, $\{A, BC\}$ can emerge as the

equilibrium in the new model. The detailed derivation and explanation are provided in Online Appendices D and E.

¹⁸ Here, $SW^{(0)}$, $SW^{(AB)}$, and $SW^{(BC)}$ denote social welfare and $CS^{(0)}$, $CS^{(AB)}$, and $CS^{(BC)}$ denote consumer surplus for structures $\{A, B, C\}$, $\{AB, C\}$, and $\{A, BC\}$, respectively. Boundaries α_i^{SW} , α_i^{CS} , k_i^{SW} , and k_i^{CS} are provided in the proof of Proposition 4.

¹⁹ For example, without efficiency gains from mergers, Salant et al. (1983) show that a merger among firms competing in quantity decreases social welfare and consumer surplus. Sinitsyn (2020), not considering capacity constraint, uses the framework of Narasimhan (1988) to model price competition and finds that a horizontal merger always decreases consumer surplus.

²⁰ For the case in which the large firm A has fewer loyal buyers than a small firm (i.e., $0 < l < 1$), we find that the mixed merger structure is the equilibrium for small α and intermediate k , and the independent structure is the equilibrium in other regions. This result is consistent with that in our base model in which $l = 1$.

²¹ Here, $\bar{k} \equiv \frac{(1-2\alpha)(1-\alpha-l\alpha)^2}{l\alpha(1-\alpha-l\alpha)+(1-2\alpha)(1-2\alpha-l\alpha)} < 1 - \alpha - l\alpha$. Also, $\bar{s}_1 \equiv (1 - 2\alpha - l\alpha)(1 - \frac{k}{1-l\alpha})$, $\bar{s}_2 \equiv \bar{s}_1(1 - \frac{k}{\alpha+\bar{s}_1})$ and $\bar{s}_3 \equiv (1 - 2\alpha - l\alpha)(1 - \frac{2k}{1-l\alpha})$, which are analogous to s_1 , s_2 and s_3 in the symmetric loyal segment setting.

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